Online Appendix to "Reconstructing the Yield Curve"*

Yan Liu,[a] and Jing Cynthia Wu[b,c†]

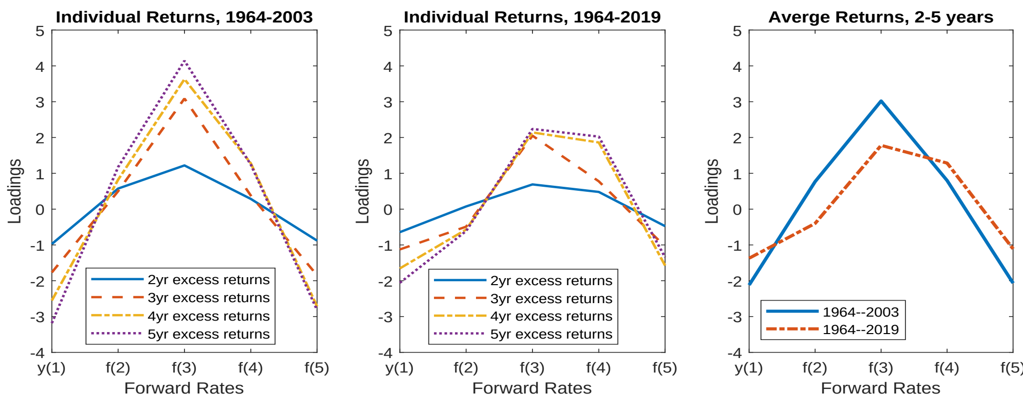
[a] *Purdue University, West Lafayette, IN 47905, USA*
[b] *University of Notre Dame, Notre Dame, IN 46556, USA*
[c] *National Bureau of Economic Research, Cambridge, MA 02138, USA*

*Our yield curve data are available at `https://sites.google.com/view/jingcynthiawu/yield-data`.

†Corresponding author.
E-mail address: liu2746@purdue.edu (Y.Liu), cynthia.wu@nd.edu (J.Wu).

Appendix D CP Regressions with Fama-Bliss Data

Fig. D.1. CP Loadings with Fama-Bliss Data



Notes: x-axis: independent variables $y_t(1), f_t(2), ..., f_t(5)$.

Table D.1. CP Regression Results with Fama-Bliss Data (Predicting 2- to 5-Year Bond Returns)

Fama-Bliss Data

		Forward Rates					PCs					
		$y(1)$	$f(2)$	$f(3)$	$f(4)$	$f(5)$	$PC1$	$PC2$	$PC3$	$PC4$	$PC5$	R-square
1964-2003	Loadings	-2.119	0.769	3.021	0.798	-2.067	0.168	-1.459	-2.624	3.143	0.437	0.351
	NW t-stat	-6.193	1.112	5.460	1.750	-4.997	2.190	-4.373	-6.237	4.981	0.570	
		(0.00)	(0.27)	(0.00)	(0.08)	(0.00)	(0.03)	(0.00)	(0.00)	(0.00)	(0.57)	
	BH 5% c.v.	4.819	2.429	2.682	2.991	2.580	4.134	6.578	2.744	2.745	2.227	
		(0.01)	(0.37)	(0.00)	(0.28)	(0.00)	(0.38)	(0.27)	(0.00)	(0.00)	(0.61)	
	F-stat									21.099		
										(0.00)		
	BH 5% c.v.									8.061		
										(0.00)		
1964-2019	Loadings	-1.371	-0.399	1.781	1.286	-1.111	0.061	-1.254	-2.188	0.933	-0.928	0.228
	NW t-stat	-3.161	-0.603	2.561	3.026	-2.385	1.043	-4.038	-4.162	1.277	-1.202	
		(0.00)	(0.55)	(0.01)	(0.00)	(0.02)	(0.30)	(0.00)	(0.00)	(0.20)	(0.23)	
	BH 5% c.v.	4.597	2.178	2.489	3.150	2.863	3.806	6.705	2.755	2.253	2.097	
		(0.25)	(0.59)	(0.04)	(0.06)	(0.10)	(0.67)	(0.40)	(0.00)	(0.26)	(0.27)	
	F-stat									4.124		
										(0.02)		
	BH 5% c.v.									5.252		
										(0.09)		

Notes: The dependent variable is the one-year excess return $\overline{rx}_{t+1}^{(2\to5)} = \frac{1}{4}\sum_{n=2}^{5} rx_{t+1}(n)$. The independent variables are the forward rates: $y_t(1), f_t(2), ..., f_t(5)$, or the five PCs ($PC1$-$PC5$) The regression has an intercept. NW t-stats and p-values are computed using Newey-West standard errors with 18 lags. F-stats and the associated p-values report the joint F test for $PC4$ and $PC5$. For both t stats and F stats, we also report the corresponding 5% critical values and p-values of the bootstrap distributions of Bauer and Hamilton (2018). p-values are reported in parentheses.

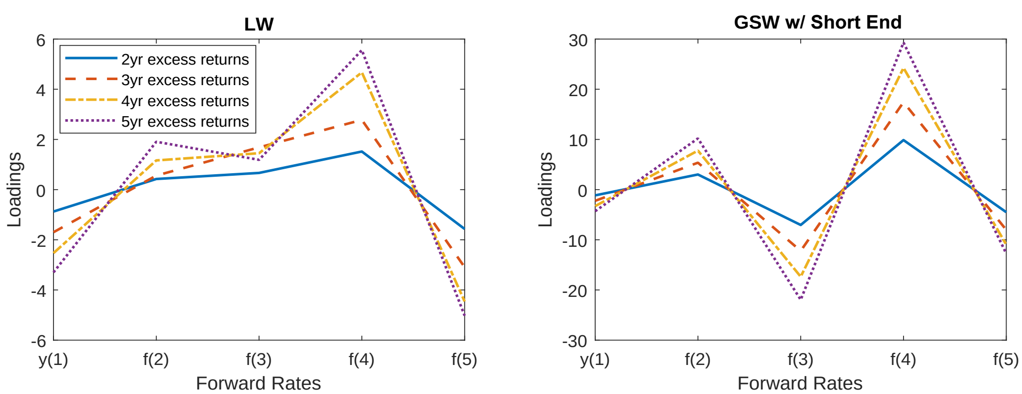
Appendix E GSW Estimated with the Short End

This appendix repeats exercises in Section 5 with one difference: we estimate the GSW curve by keeping the securities with short maturities and Treasury bills.

E.1 *Cochrane-Piazzesi Return-Forecasting Regressions: GSW Estimated with the Short End*

With the GSW data that is estimated with the short end, loadings in the right panel of Figure E.1 have a similar pattern across different maturities. However, the magnitude is even more extreme than what's in Figure 4.

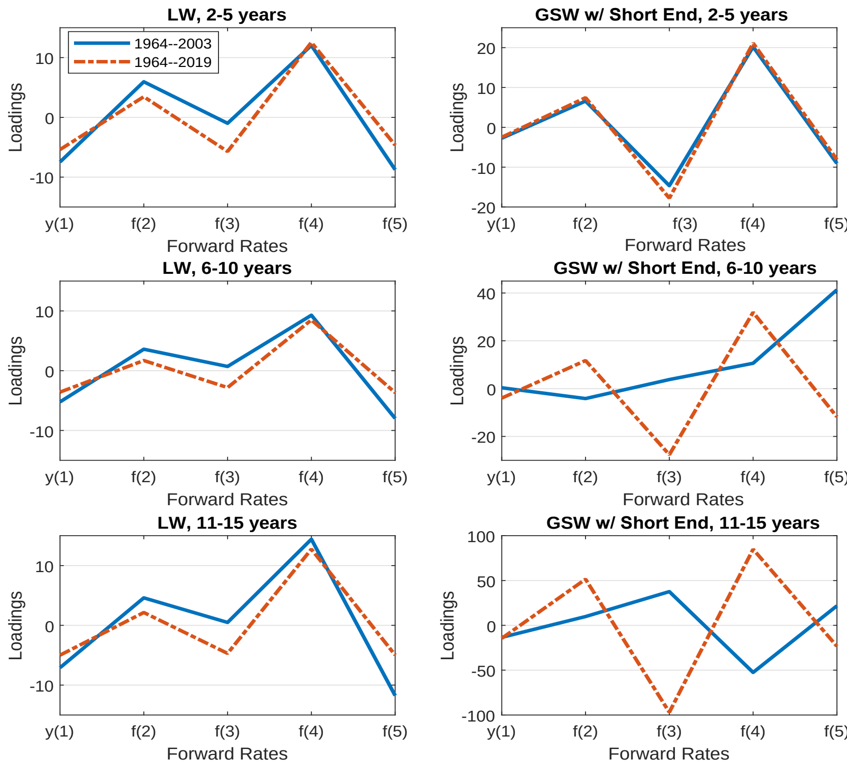
Fig. E.1. CP Loadings of Individual Excess Returns



Notes: This figure repeats Figure 4 except that the GSW curve is estimated with the raw data in the short end. x-axis: independent variables $y_t(1), f_t(2), ..., f_t(5)$. Different lines represent excess returns with different maturities $rx_{t+1}(2), ..., rx_{t+1}(5)$. We use CP's original sample from 1964-2003.

Moreover, the shapes of GSW loadings change once we extend the maturity beyond five years, and they look different between CP's original sample (1964-2003) and the extended sample (1964-2019); see the right panels of Figure E.2. These results are consistent with Figure 5.

Fig. E.2. CP Loadings of Average Excess Returns

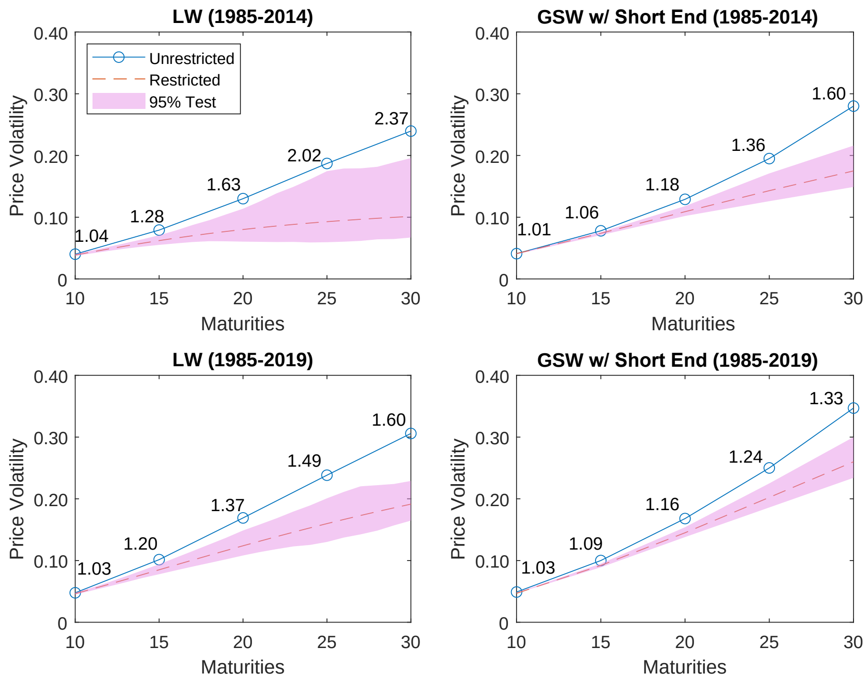


Notes: This figure repeats Figure 5 except that the GSW curve is estimated with the raw data in the short end. x-axis: independent variables one- to five-year forward rates $y_t(1), f_t(2), ..., f_t(5)$. Dependent variables are average excess returns. Bond maturities in the top panels are 2 to 5 years, 6 to 10 years in the middle panels, and 11 to 15 years in the bottom panels. Blue lines are CP's original sample from 1964-2003; red lines in the middle and bottom panels are the extended sample from 1964-2019.

E.2 *Excess Volatility: GSW Estimated with the Short End*

Figure E.3 confirms the results in Figure 6.

Fig. E.3. Excess Volatility

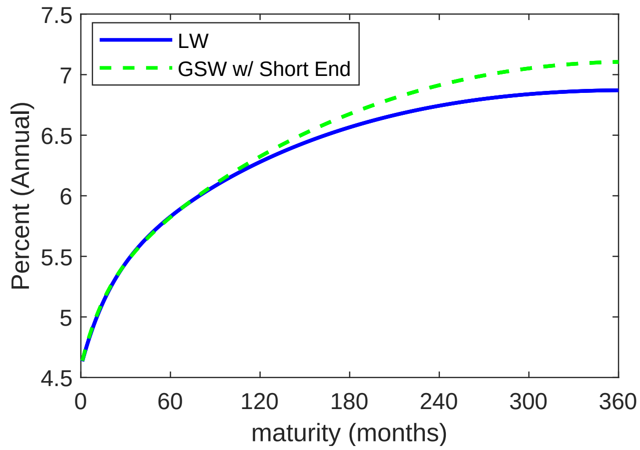


Notes: This figure repeats Figure 6 except that the GSW curve is estimated with the raw data in the short end. We plot the unrestricted variance $V^u(n)$ (solid line) and the restricted variance $V^r(n)$ (dashed line) for the log price of long-term bonds. The shaded area marks the 95% confidence bands. The circles highlight the variance ratios $\frac{V^u(n)}{V^r(n)}$ for bonds with selected maturities.

E.3 *Gaussian Affine Term-Structure Model: GSW Estimated with the Short End*

For the term-structure model, the differences lie in the long end. The difference of unconditional yield curves is about 0.3% (annualized) in the long end, which is large; see Figure E.4.

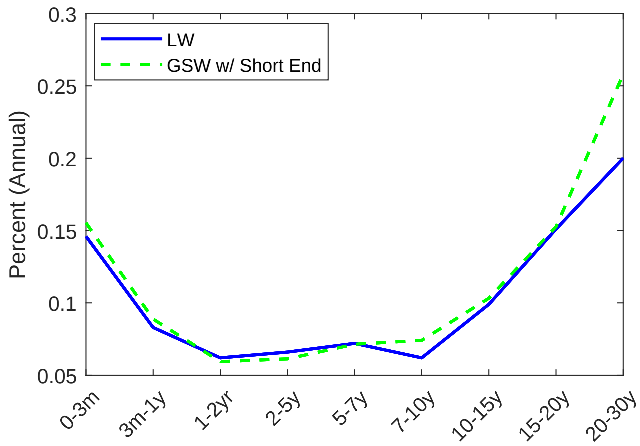
Fig. E.4. Unconditional Yield Curve



Notes: This figure repeats Figure 7 except that the GSW curve is estimated with the raw data in the short end. Cross section of yield curve implied by the term-structure model fitted to the LW (in the blue solid line) and GSW (in the green dashed line) zero-coupon yields.

Comparing Figure E.5 with Figure 8, the difference in pricing errors is now larger in the long end, with that of GSW being over 0.05% higher.

Fig. E.5. Pricing errors



Notes: This figure repeats Figure 8 except that the GSW curve is estimated with the raw data in the short end. Pricing errors calculated as mean absolute yield error in annualized percentage points. Blue: use LW zero-coupon yields to fit the term-structure model, and errors are calculated based on the raw coupon-bearing Treasury securities and the fitted yield curve; Green: use GSW zero-coupon yields to fit the term-structure model, and errors are calculated based on the raw coupon-bearing Treasury securities and the fitted yield curve.